

# Cindy Hsu

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Portfolio: <https://cindyhsugit.github.io/playground/>

LinkedIn: <https://www.linkedin.com/in/cindy-hsu-80b20928/>

GitHub: <https://github.com/cindyhsugit/ai-chatbot-fastapi-with-rag>

## Professional Summary

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Junior AI/ML engineer with hands-on, production-grade experience building retrieval-augmented generation (RAG) systems. Designed and shipped a two-stage retrieval pipeline (FAISS + HuggingFace cross-encoder reranking), a provider-agnostic generation layer (OpenAI and Gemini) with automatic failover, and self-directed adversarial testing to verify system correctness. Brings 12+ years of production software engineering discipline — version control, testing, deployment, incident debugging — to a deliberate, focused transition into AI/ML engineering.

*Career note: 2019–2025, extended personal sabbatical (travel); 2025–present, self-directed transition into applied AI/ML engineering.*

## Core Skills

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Python, FastAPI, RAG pipeline design, FAISS, HuggingFace (sentence-transformers, cross-encoder reranking), OpenAI API, Gemini API, multi-provider LLM integration, retry logic & exponential backoff, async/await, prompt engineering, Docker, Git & GitHub, REST APIs, pytest, cloud deployment (Render, Google Cloud Run), environment variable & secrets management, JSON, command line.

## Projects

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### AI Chatbot with RAG — Python, FastAPI, FAISS, HuggingFace, OpenAI API, Gemini API, Docker

Code: <https://github.com/cindyhsugit/ai-chatbot-fastapi-with-rag>

- Built and deployed a containerized FastAPI chatbot, live on two independent platforms (Render and Google Cloud Run), each requiring adaptation to its own deployment model.
- Designed a two-stage retrieval architecture: FAISS wide-recall retrieval (top 20 candidates) followed by HuggingFace cross-encoder reranking (ms-marco-MiniLM-L-6-v2) narrowed to the top 3–5, improving answer relevance beyond raw vector similarity alone.
- Migrated embeddings from OpenAI's paid API to a self-hosted HuggingFace model (all-MiniLM-L6-v2), removing per-call cost and external dependency; diagnosed and resolved a resulting FAISS dimensionality mismatch (1536 vs. 384) caused by inconsistent embedding usage across synchronous and asynchronous code paths.
- Built a provider-agnostic generation layer supporting OpenAI and Gemini; implemented retry logic with exponential backoff and automatic failover between providers, validated under a real production API outage.
- Verified retrieval correctness through self-directed adversarial testing — injecting fabricated facts into source documents to confirm answers came from retrieved context, not model pretraining.
- Resolved real cross-environment issues: a Python version mismatch breaking the Docker build, missing dependencies, and secure credential handling across local, containerized, and production environments.
- Version-controlled with Git across multiple feature branches, including resolving a diverged branch and merge conflict during a history cleanup; wrote unit/integration tests for API endpoints using pytest and TestClient.

## Prior Professional Experience — Software Engineering

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### Senior iOS Developer

StubHub, San Francisco | March 2017 – May 2019

- Developed new features for different iOS versions, share extensions, widgets, and public/private libraries.
- Enhanced UI behavior for video/audio call states and responsive settings across iPad and iPhone.
- Improved localization support, increasing non-English speaking users by 300%.

### Senior iOS Developer

Udemy, San Francisco | April 2015 – February 2017

- Handled build and release processes while collaborating with QA overseas.
- Translated the codebase to Swift and increased Swift usage across the project.

### Senior iOS Developer

Mercari, San Francisco | September 2014 – April 2015

- Worked with the Japan team to maintain code consistency and support new feature development.
- Tested U.S. and Japan app versions and supported localization testing in English and Japanese.

## Education

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M.S. Computer Science — University of Southern California

B.S. Computer Science — Iowa State University